Skills Summary

Measuring and Picture Graphs

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a picture graph (read-picture-graph)

Rule:

A picture graph answers a question with *pictures you count.* Always read the **key** first — it tells you what one picture is worth. Then find the **one row** the question names and touch each picture once as you count it.

Pet Show: Cats $\star\star\star\star$ Dogs $\star\star\star\star\star\star\star$ Birds $\star\star$ (each $\star$ = 1 pet)

Example: How many *dogs* are in the Pet Show graph?
Step 1. The question names one row — dogs — so I count only that row, not the whole graph.
Step 2. Touch and count the dog stars: 1, 2, 3, 4, 5, 6, 7. (Cats show 4 and birds show 2, but those rows were not asked about.) $\Rightarrow$ number of dogs = **7**

Try it: Marble jar: Red $\star\star\star\star\star$ Blue $\star\star\star$ Green $\star\star\star\star\star\star$. How many *green* marbles?

check: **6**

Skill 2 — Measuring with same-size units

Rule:

Lay the units end to end with **no gaps and no overlaps**, starting at one edge of the object. The number of units is the length.

Two lengths can be compared by their numbers **only when the same unit was used both times**.

Example: A ribbon is 8 cubes long. A pencil is 5 cubes long. How many cubes longer is the ribbon?
Step 1. Both were measured with the same cube, so "how many longer" is just the gap between the two counts — subtract.
Step 2. Count up from 5 to 8: 6, 7, 8 — three counts. So $8 - 5 = 3$. $\Rightarrow$ the ribbon is **3** cubes longer

Try it: A straw is 10 cubes long and a marker is 6 cubes long. How many cubes longer is the straw?

check: **4**

Skill 3 — Comparing from data (compare-from-data)

Rule:

"How many more", "who has the fewest", "is that true" — all of these need **both** amounts first. Get each number from the graph, *then* compare.

$>$ and $<$ always open toward the bigger amount: $7 > 4$ and $4 < 7$.

Example: In the Pet Show graph, how many *more* dogs than cats?

Step 1. "How many more" compares two rows, so I read both counts before doing anything else.

Step 2. Dogs 7, cats 4. Match them up one to one; the leftover dogs are the extra: $7 - 4 = 3$.

$\Rightarrow$ there are **3** more dogs than cats

Try it: Marble jar again (Red 5, Blue 3, Green 6). How many *more* green marbles than blue?

check: **3**

(!) Watch out: A bigger *count* does not always mean a bigger *unit*. If the same straw is 9 paper clips but only 5 blocks, the block is the longer unit — it takes fewer of a long unit to cover the same length. And when a question asks "how many more", do not stop at the bigger row: the answer is the difference, not the total.